NLP Course (097215) · Spring 2026

Final Project

Text-Tabular Learning

AT A GLANCE

Deadline	**Monday, October 26, 2026** (all deliverables)
Submission	Individual work (no pairs), via the course Moodle
Tracks	Choose **one**: Track 1 (Algorithm) or Track 2 (Benchmark)
Key lecture	July 19, 2026: MulTaBench, presented by Alan Arazi
Final report	Short paper in English, up to 4 pages, NeurIPS template
AI tools	Permitted and encouraged; you are responsible for everything you submit

1 Overview & Background

In many real-world applications, data does not arrive in a single modality. Instead, it is often structured in tables (tabular data) where some columns contain standard numerical or categorical features (e.g., a patient's age or blood type), while other columns contain free-text descriptions, medical notes, or product reviews. This setting is known as **text-tabular learning** (a sub-field of Multimodal Tabular Learning).

Standard tabular algorithms, such as Gradient Boosted Decision Trees (GBDTs like XGBoost [9], CatBoost [7], and LightGBM [8]) or Tabular Foundation Models (TFMs like TabPFN [3, 4] and TabICL [10, 11]), are natively unimodal and cannot process raw text. To solve text-tabular problems, practitioners must either:

1. **Represent the text via static embeddings:** Use a pretrained text encoder (e.g., E5 [12] or RoBERTa [14]) to generate embeddings, project/compress them, and append them as numerical columns alongside the structured features.
2. **Build joint models:** Train a dedicated architecture that natively fuses text representations with tabular features. Some joint models additionally allow the text encoder to adapt to the prediction target via **Target-Aware Representations (TAR)**, as in TabSTAR [2].

On **July 19th**, we will have a dedicated lecture on text-tabular prediction problems. During this lecture, **Alan Arazi** will present the MulTaBench [1] paper, which introduces a curation pipeline and benchmark for evaluating multimodal tabular learning models.

1.1 Choosing Your Track

For your final project, you can choose between two main tracks:

	Track 1: Algorithm (§2)	Track 2: Benchmark (§3)
Your mission	Design an ML algorithm for prediction tasks over tabular data with free-text features	Curate a new text-tabular dataset that passes the MulTaBench curation pipeline
You submit	Source code, evaluation results, short paper	Dataset(s) CSV(s), source code, evaluation report, short paper
Success means	Statistically significant improvement over the relevant MulTaBench baseline	Passing both curation criteria: joint signal and TAR gain

1.2 Advising & Support

The project advisor is **Eilam Shapira**. A dedicated forum will be opened in the course Moodle for all project-related questions. We will do our best to answer every message posted in the forum within a week, including during the summer break and the holidays.

> **AI USAGE POLICY**
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> The use of any kind of artificial intelligence tool is permitted, and even encouraged. Note, however, that you are responsible for **everything** you submit.

2 Track 1: The Algorithm Track

2.1 Objective

In this track, your goal is to design and implement a machine learning algorithm capable of handling classification and/or regression tasks on tabular datasets that include free-text features.

Your algorithm will be evaluated on the datasets from the MulTaBench [1] benchmark. The primary objective depends on your architectural choice:

- If your algorithm **does not** use the target variable to build its text representations, your goal is to demonstrate a statistically significant performance improvement over the **Joint Frozen** baseline.
- If your algorithm **does** use the target variable to build its text representations, whether through encoder fine-tuning or in any other way, your goal is to outperform the **Joint TAR (Target-Aware Representation)** baseline.

> **CREATIVITY MATTERS**
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> Alongside the quantitative objective above, creativity matters too: a creative, well-motivated

> approach, backed by a deep and thorough analysis, will be highly valued even if it does not decisively outperform the baseline. Conversely, a trivial modification, such as simply swapping the text encoder for a stronger embedding model, is not an interesting contribution on its own, even if it improves the score.

2.2 Project Ideas & Directions

- **Tabular-Aware Text Encoders:** Modify a text encoder (or its input verbalization) to dynamically condition on numerical or categorical tabular metadata.
- **Smart Embedding Fusion:** Design novel fusion architectures to combine text representations (e.g., from the E5 model family [12]) with tabular features more effectively.
- **Data Augmentation:** Implement a method that generates realistic synthetic text-tabular samples to augment the training set and improve downstream model performance.

> **DATA LEAKAGE PREVENTION**
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> You must ensure that your training or pretraining phase does not leak test set information. If you use data from a MulTaBench dataset during training/fine-tuning, you must evaluate that specific dataset using a model variant that has *never* seen it during its training process. For instance, you should train separate models (or use cross-validation) to evaluate each dataset in the benchmark cleanly.

2.3 Submission Requirements

1. **Source Code:** A general-purpose implementation of your algorithm that can be executed on any arbitrary text-tabular dataset. Package it with pinned dependencies (e.g., a `pyproject.toml` with a `uv.lock`, or an equivalent convention) so that we can run your code with exactly the packages you used.
2. **Evaluation Results:** Performance reports (specifically, AUROC for classification and R^2 for regression) across all datasets in the MulTaBench benchmark.
3. **Pretrained Weights & Hugging Face:** If your code utilizes a pretrained/fine-tuned model:
   - Upload the model weights to **Hugging Face**. Your code can load them directly from Hugging Face; public models are downloaded automatically, so no dedicated download script is needed.
   - Submit two versions of your code:
     a. **Inference Mode:** The algorithm running with your uploaded model(s).
     b. **End-to-End Training Mode:** A script that trains/fine-tunes the model(s) from scratch, reproducing the uploaded weights.
4. **Final Report (Short Paper):** A short academic paper describing your project, as detailed in Section 4.

3 Track 2: The Benchmark Track

3.1 Objective

In this track, your task is to identify, preprocess, and evaluate a new dataset to determine if it satisfies the curation criteria of a text-tabular learning task. You will analyze your dataset's properties by applying the curation pipeline developed in the MulTaBench [1] study.

3.2 MulTaBench Curation Pipeline

To verify if a text-tabular dataset carries the necessary signal, you must evaluate it across **5 diverse tabular learners**: TabM [6], CatBoost [7], LightGBM [8], TabPFN 2.0 [4], and TabPFN 2.5 [5]. To run this evaluation, use the official MulTaBench source code, which already implements all five learners and all four experimental conditions described below.

For each learner, evaluate the performance under **4 distinct conditions**:

- **Unimodal Structured:** Tabular features only (numerical and categorical).
- **Unimodal Unstructured:** Text features only.
- **Joint Frozen:** All features combined, using static, frozen text embeddings (e.g., E5-small-v2 [12]).
- **Joint Target-Aware (TAR):** All features combined, where the text encoder has been fine-tuned (e.g., using LoRA [13]) on the target variable.

3.2.1 Curation Criteria

For a dataset to successfully pass the curation pipeline, it must satisfy both of the following conditions for **at least 3 out of the 5** tabular learners:

1. **Joint Signal:** The performance under the *Joint Frozen* condition must be strictly higher than both the *Unimodal Structured* and *Unimodal Unstructured* baselines. This ensures that both modalities contain complementary predictive signals and prevents the dataset from collapsing into a pure NLP task.
2. **Task-Awareness (TAR Gain):** The performance under the *Joint TAR* condition must be strictly higher than the *Joint Frozen* condition. This isolates the empirical benefit of target-aware representation tuning.

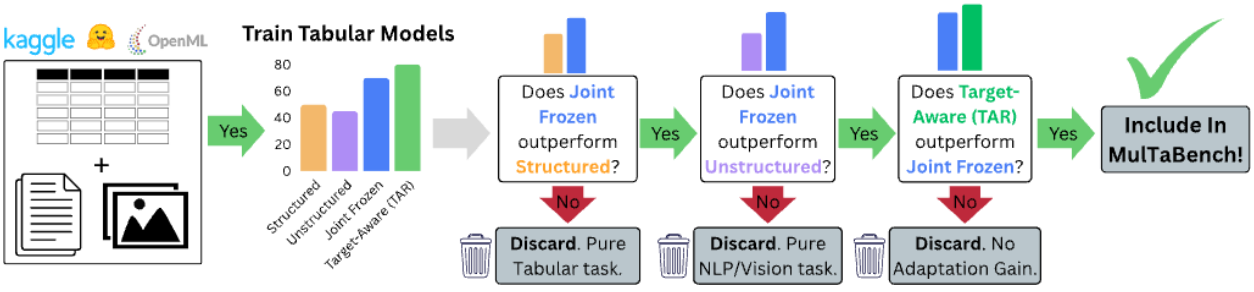


Figure 1: The MulTaBench Curation Pipeline. Datasets are included if joint prediction outperforms unimodal baselines and if Target-Aware Representations (TAR) improve on frozen, off-the-shelf embeddings.

DATASET EVALUATION & GRADING

The magnitude of the performance delta between the conditions will be the primary metric used to evaluate and grade the quality of your dataset. A larger, robust performance gap demonstrates a more successful and challenging curation process, which will reflect positively on your final project grade.

FUTURE COLLABORATION OPPORTUNITY

In the future, the course staff may collect the highest-quality datasets compiled by students in this track to build a larger, combined text-tabular benchmark. If a dataset you curate is included in this future benchmark, you will be invited to join as a co-author on the resulting academic publication.

3.3 Project Scope and Grading Tracks

To help guide your work, the final project can be completed at two different levels of scope:

3.3.1 Standard Project (Default Target)

The standard requirement is to identify, preprocess, and evaluate **one** new text-tabular dataset that meets the MulTaBench curation pipeline criteria. You must present evaluation results showing it satisfies the joint signal and task-awareness conditions.

- *Ideas for Standard Projects:*
  - Clean and restructure a raw public dataset (e.g., from Kaggle or OpenML) that contains text fields but is not currently part of MulTaBench.
  - Inject informative text features into a structured tabular dataset.
  - Incorporate structured features (e.g., metadata, user stats) into an NLP dataset.

3.3.2 Outstanding Project (Optional Extension)

For students looking for an extra challenge, you may propose an outstanding project. This requires developing a systematic method to automatically generate or mine new text-tabular datasets, and submitting at least **5 new datasets** generated or discovered via your method that successfully pass the MulTaBench curation criteria.

- *Ideas for Outstanding Projects:*
  - **Tabular-to-Text Enrichment:** Build a general pipeline that automatically enriches numerical/categorical tables with relevant text descriptions mined from the web.
  - **Generative Tabular Synthesis:** Use Large Language Models (LLMs) to synthesize realistic text-tabular datasets from scratch.
  - **Automated Dataset Mining:** Build a web crawler that automatically discovers raw text-tabular data sources, parses them, and checks if they pass the curation filters.
  - **Image-Text-Tabular Datasets:** Curate datasets that combine all three modalities (images, text, and structured features) and pass the curation criteria for *all* modalities, as in the

PetFinder analysis in the MulTaBench paper [1]. Such datasets are especially valuable and very hard to create.

> **SCOPE GUIDANCE**
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> We strongly recommend that all students target the **Standard Project** as their starting point. Do not attempt to design or plan for the *Outstanding Project* from the beginning. You should only expand your scope if you find yourself highly motivated, are enjoying the project, and have ample time remaining. Grading will reflect the scope and execution of the project accordingly (the Standard Project is fully sufficient for a top grade, while the Outstanding Project offers potential extra credit).

3.4 Submission Requirements

1. **Dataset Files & Code:** The curated dataset(s) in CSV format, together with the code used to create and evaluate them.
2. **Evaluation Report:** A CSV file detailing the performance of the 5 models across the 4 experimental conditions for each dataset.
3. **Final Report (Short Paper):** A short academic paper describing your project, as detailed in Section 4.

4 Final Report Guidelines (Short Paper)

Regardless of the track chosen, each student must submit a short academic paper written in English.

- **Format:** NeurIPS LaTeX template.
- **Length:** Up to **4 pages** (excluding references and appendices).
- **Appendices:** Restricted to result tables/metrics only.

4.1 Required Sections

1. **Title & Abstract**
2. **Introduction:** Briefly describe the problem, the context of text-tabular learning, and what your project achieves.
3. **Related Work:** Review and cite relevant literature, including the papers behind the models you used and other relevant work (e.g., MulTaBench).
4. **Method / Dataset:**
   - *For Track 1:* Detail the architecture and design of your proposed algorithm.
   - *For Track 2:* Detail your data curation methodology, dataset origins, and pipeline.
5. **Results:** Present your experimental setup, baselines, and performance metrics (use tables and charts).
6. **Conclusion & Summary:** Summarize the key outcomes, discuss limitations, and propose directions for future work.

SUBMISSION & DEADLINE

The project is completed **individually** (no pairs). All deliverables, for both tracks, are submitted via the course **Moodle** and are due by **Monday, October 26, 2026**. Late submissions incur a penalty of **20 points** off the final grade for each day of delay not covered by the regulations of the Technion's School of Undergraduate Studies.

References

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